IC103 Lecture 3: Key Takeaways

Costs, Revenue and Profit

1. Short-Run Costs

- **Explicit costs:** actual payments such as wages, rent and electricity.
- **Implicit costs:** opportunity costs of using your own resources.
- **Sunk costs:** costs already incurred and cannot be recovered.
- **Fixed costs** do not change with output.
- **Variable costs** change with output.

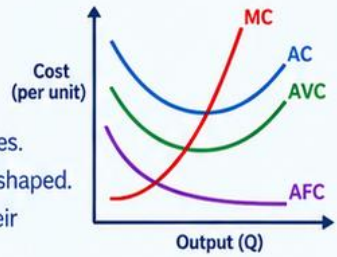


$$TC = TFC + TVC$$

In the short run, some costs are fixed and some vary with output.

2. Short-Run Cost Curves

- TFC is constant.
- TVC rises as output rises.
- TC rises because variable cost rises.
- AFC falls as output increases.
- AVC and AC are usually U-shaped.
- MC cuts AVC and AC at their minimum points.



Marginal cost helps explain the shape of average cost curves.

3. Production and Cost

- As marginal product rises, marginal cost falls.
- As diminishing marginal returns set in, marginal product falls.
- When marginal product falls, marginal cost rises.
- This links production theory to cost theory.



Diminishing marginal returns lead to rising marginal cost.

4. Long Run and Returns to Scale

- In the long run, all inputs are variable.
- Firms can change their scale of production.
- **Increasing returns to scale:** output rises by more than inputs.
- **Constant returns to scale:** output rises in the same proportion as inputs.
- **Decreasing returns to scale:** output rises by less than inputs.



Returns to scale are a long-run concept because all inputs can change.

5. Economies of Scale and Scope

Scale

- Producing more of the same product lowers average cost.
- **Reasons:** specialisation, large machinery, indivisibilities, finance, management.



Scope

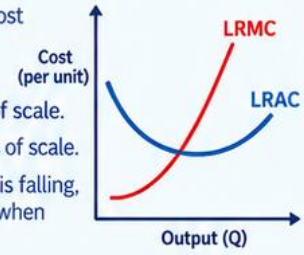
- Producing different products together lowers cost.
- **Reasons:** shared management, marketing, buildings, equipment or technology.



Scale = more of one product. Scope = different products together.

6. Long-Run Cost Curves

- LRAC shows the lowest average cost for each output level.
- LRAC may fall, flatten, then rise.
- Falling LRAC reflects economies of scale.
- Rising LRAC reflects diseconomies of scale.
- LRMC is below LRAC when LRAC is falling, equal at the minimum, and above when LRAC is rising.



The long-run average cost curve shows efficient scale choices.

7. Revenue Concepts

- **Total revenue (TR)** = price × quantity sold.
- **Average revenue (AR)** = TR / Q.
- **Marginal revenue (MR)** = extra revenue from selling one more unit.
- In perfect competition, the firm is a price taker.
- For a price-taking firm: P = AR = MR.

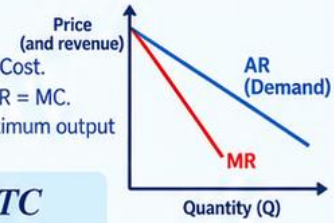


$$TR = P \times Q \qquad AR = \frac{TR}{Q}$$

A price-taking firm faces a horizontal demand curve at the market price.

8. Revenue in Other Markets and Profit Maximisation

- In imperfect competition, the demand curve slopes downward.
- AR is the demand curve.
- MR lies below AR.
- Profit = Total Revenue – Total Cost.
- Firms maximise profit where MR = MC.
- This does not always mean maximum output or revenue.



$$Profit = TR - TC$$

Profit is maximised where marginal revenue equals marginal cost.

Quick Revision Checklist

- ☐ Distinguish fixed, variable and sunk costs.
- ☐ Explain the shapes of cost curves.
- ☐ Link marginal product to marginal cost.
- ☐ Define returns to scale.
- ☐ Distinguish economies of scale from economies of scope.
- ☐ Define TR, AR and MR.
- ☐ Explain why P = AR = MR for a price-taking firm.
- ☐ State the profit-maximising rule MR = MC.